



Department of Computer Science and Business Systems Academic Year 2023 – 2024(Even Semester)

Degree, Semester & Branch: B.Tech, IV & CSBS

Course Code & Title: AL3451 Machine Learning

Name of the Faculty member: Mrs.B.Yazhini AP/CSBS

Innovative Practice Description

- **Unit / Topic:** Unit III / Ensemble Learning-Bagging, Boosting
- **Course Outcome:** CO3 –Characterize the machine learning algorithms as unsupervised learning and analyze the various algorithms of ensemble techniques and unsupervised learning
- **Topic Learning Outcome:** TLO 4,5
- **Activity Chosen:** Peer Teaching
- **Justification:**
 - Peer Teaching fosters teamwork, cooperation, patience, and better social skills. In a cooperative peer learning environment, each student's strengths can serve to complement the group and enhance learning. Teaching by their peers will improve the student's understanding.
- **Time Allotted for the Activity:** 30 Minutes
- **Details of the Implementation:**
 1. **Materials used:**
 - a. Christopher M. Bishop, "Pattern Recognition and Machine Learning", Springer, 2006.
 2. **Prerequisite:** Basic knowledge on Machine Learning and Ensemble Learning
 3. All students were asked to discuss in their teams on the given topic with the help of textbook.
 4. 10 Interested students are asked to explain the concepts to their class members.
 5. To check their knowledge on this topic, questions were asked to students.
 6. After that, the faculty consolidated the concepts once again to students and the answers for the questions were also discussed.

• **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO4	PO5	PO9	PO12	PSO1	PSO3
CO3	3	3	3	2	2	2	2	3	2

(1 – Low 2 – Moderate 3 – High)

Innovative practice	PO1 3	PO2 3	PO3 3	PO4 2	PO5 2	PO9 2
Justification for correlation	Students will be able to apply supervised learning algorithms for making predictions	Students can be able to identify and analyze the problem	Students can be able to design supervised machine learning algorithms for solving the problem	Students will apply research-based knowledge and methodologies to validate the effectiveness and performance of various Machine learning algorithms	Students will deploy the linear regression, classification and generative models.	Students will work effectively as individuals and as members of teams to develop and deploy machine learning models in real world scenario

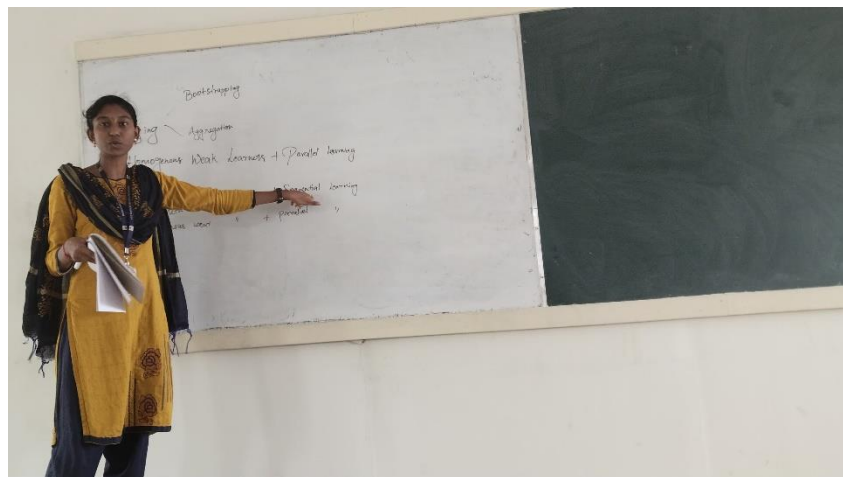
PO mapped:

Innovative practice	PO11 2	PO12 2
Justification for correlation	Students will manage projects related to Decision Tree Regression, Naïve Bayes and Support Vector machine	Students will be encouraged to explore emerging Machine learning application through continuous learning and research

PSO Mapped:

Innovative practice	PSO1 3	PSO3 2
Justification for correlation	Students will manage projects related to Decision Tree Regression, Naïve Bayes and Support Vector machine	Students will be encouraged to explore emerging Machine learning application through continuous learning and research

- **Images / Screenshot of the practice:**



Sample screenshots of peer learning

- **Reflective Critique:**

- ❖ ***Feedback of practice from students and other stakeholders:***

Students felt that they can able to understand and get clarity on the topic very easily by sharing their knowledge with their friends. It improves their self-learning ability.

- ❖ ***Benefit of the practice:***

Students were actively participated in this activity.

Students interaction with their peer members and the class during the activity was good and effective.

- ❖ ***Challenges faced in implementation:***

Only few students are ready to present the topic in front of their class. Some students are hesitated to share their knowledge in the class. And they are motivated to come out of their fear.

References:

1. <https://www.wgu.edu/blog/peer-learning2208.html#close>
2. https://www.ritrjpm.ac.in/images/CSBS/2023-2024/PT_AD3351_DAA_KU_23_24.pdf